

TECHNICAL REPORT v1.0

Smart Infusion Pump Software Simulator

Comprehensive analysis of system design, requirements, architecture, and validation for high-fidelity clinical training.

PLATFORM

Flutter / Supabase

SECURITY

RBAC / AES-256

EXECUTIVE SUMMARY

System Purpose

The Smart Infusion Pump Simulator is an educational mobile application designed to replicate real-world pump behavior in a risk-free environment. It focuses on safety limits, alarm management, and clinical governance.

Core Value Proposition

-  **Zero Risk:** Training without physical patient harm.
-  **Role Separation:** Precise UX for Doctors, Nurses, and Admins.
-  **Real-time Engine:** Accurate flow simulation on-device.

Technical Foundation

Built using Flutter 3.x for cross-platform availability, leveraging Supabase for secure cloud synchronization and Realtime alarm broadcasting.

FSM
Simulation

PostgreSQL
Backend

Riverpod
State

AES-256
Encryption

PHASE 01

Requirements Analysis

Functional, Non-Functional, Security, and Safety Guardrails

FUNCTIONAL REQUIREMENTS

Doctor

- Manage Drug Library protocols.
- Define mandatory hard/soft limits.
- Generate PDF Infusion Reports.
- Audit immutable session logs.

Nurse

- Select drug & program parameters.
- Operate Start/Pause/Stop controls.
- Respond to Real-time alarms.
- Monitor Battery & VTBI progress.

Admin

- Manage RBAC User Accounts.
- Monitor System Health metrics.
- Review Technical Audit Trails.
- Global System Configuration.

**Strict separation ensures clinical governance remains distinct from pump operation.*

NON-FUNCTIONAL & SECURITY

System Performance

Metric	Target
Dashboard Refresh Rate	1.0 Hz (1s)
UI Update Latency	< 100ms
Control Response	< 500ms
PDF Report Gen	< 3s (1000 entries)



Security Architecture

- **RBAC:** Role-Based Access Control enforced at Database level (RLS).
- **Encryption:** TLS for transit; AES-256 for secure storage at rest.
- **Auth:** Supabase Auth with bcrypt hashing; Biometric support.
- **Immutability:** Audit logs permit only INSERT; no DELETE/UPDATE possible.

SAFETY REQUIREMENTS

⚠️ Guardrail Logic

Safety is maintained via a tiered limit system enforced by the local simulation engine:

- **Hard Limits:** Absolute ceiling. Violations trigger immediate input rejection and a blocking modal.
- **Soft Limits:** Advisory thresholds. Triggers a yellow warning banner requiring active nurse acknowledgement.
- **Emergency Stop:** A physical-emulation button requiring a 2-second hold to prevent accidental triggers.
- **Battery Failsafe:** Auto-stop and critical alarm at 5% capacity.

ON



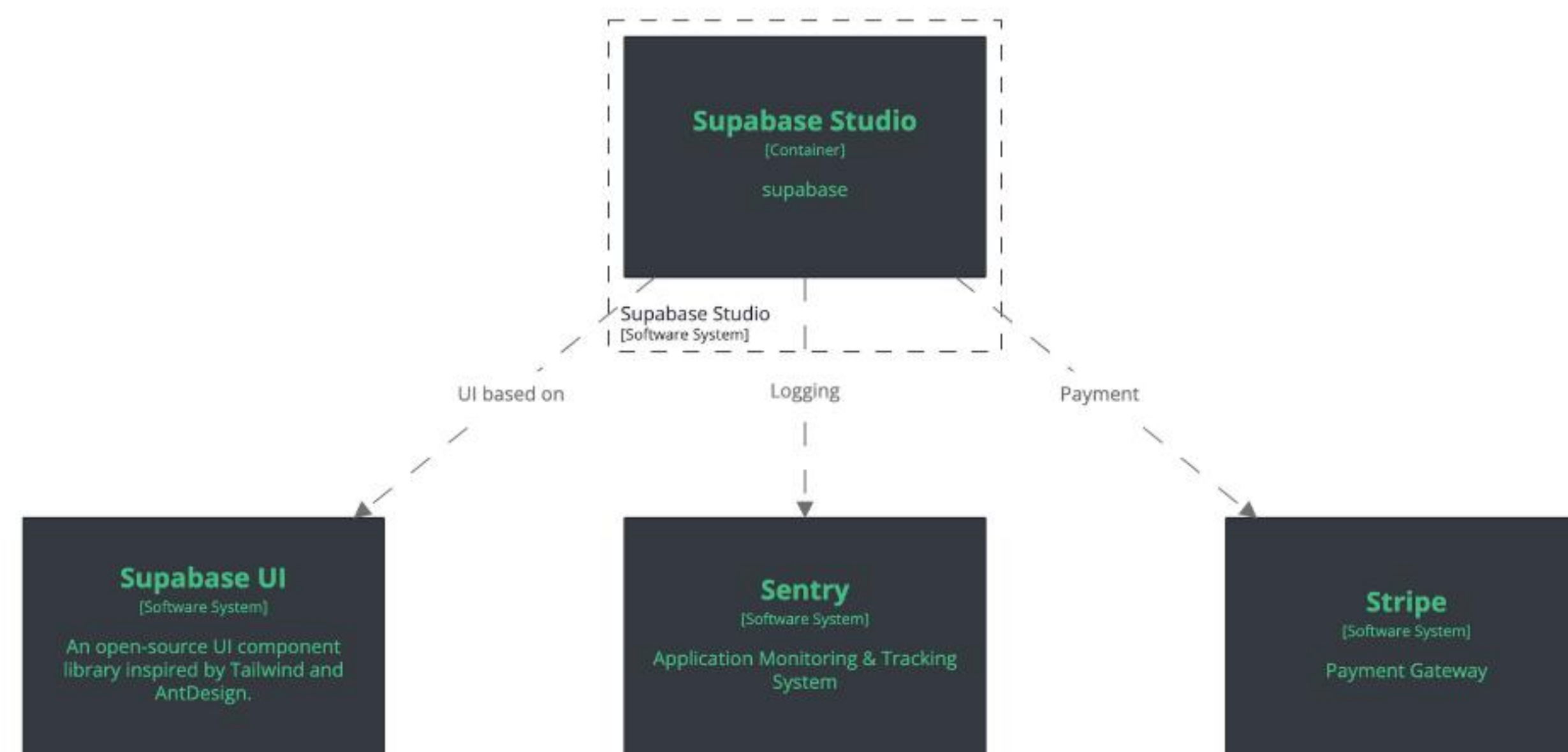
PHASE 02

System Architecture



Full-Stack Framework and Development Paradigm

TECHNOLOGY STACK



[Container] Supabase Studio
app.supabase.com
Tuesday, 5 July 2022 at 12:34 Central European Summer Time

Development Frameworks

- **UI Layer:** Flutter 3.x for declarative, reactive interfaces.
- **State Management:** Riverpod 2.5 (Compile-safe, provider-based).
- **Navigation:** GoRouter 14.x for deep-linkable, role-guarded routes.
- **Backend-as-a-Service:** Supabase (Auth, PostgreSQL, Realtime).
- **Simulation:** Local Dart-based FSM with Timer-loop.

ARCHITECTURE DETAIL

Step-by-Step Development

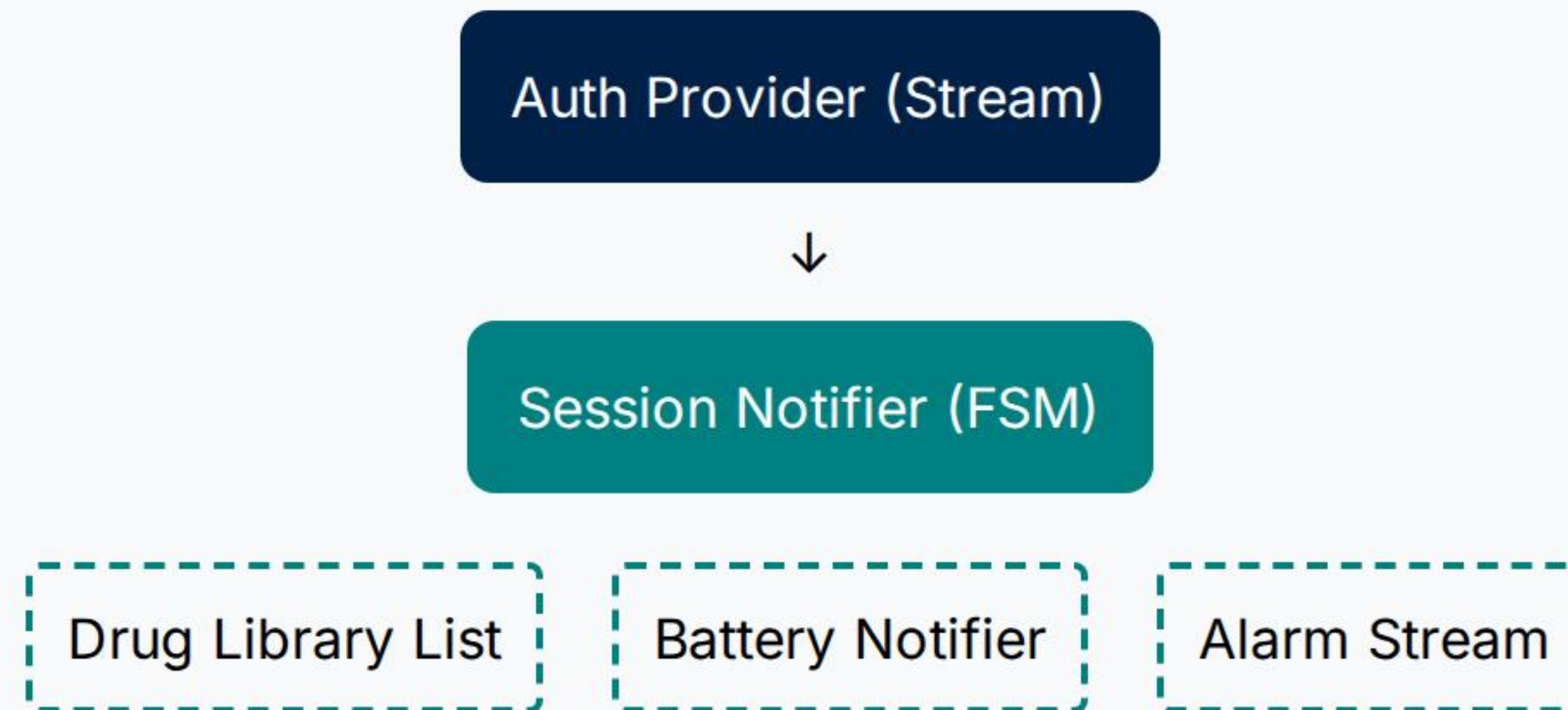
1. Auth Shell & Role Router
2. Local Simulation Engine (FSM)
3. Nurse Dashboard & Controls
4. Doctor Drug Library & Editor
5. Supabase Realtime Integration

Logical Folder Structure

```
lib/  
├── core/ (Router, Theme, Utils)  
├── features/  
│   ├── auth/ (Login, AuthRepo)  
│   ├── nurse/ (Simulation, Dashboard)  
│   ├── doctor/ (DrugLib, Reports)  
│   └── admin/ (Users, SystemHealth)  
└── shared/ (Widgets, Session)
```


STATE MANAGEMENT STRATEGY

Riverpod Provider Hierarchy



The **InfusionNotifier** acts as the central Finite State Machine (FSM), managing transitions between *Idle*, *Programming*, *Running*, *Paused*, *Alarm*, and *Complete*. This ensures UI consistency across all devices.

PHASE 03

Design & UI/UX Overview



Interface Principles and Human-Centered Simulation

| UX PHILOSOPHY

Strict Role Isolation

To eliminate visual clutter and cognitive load, the system uses a strictly segregated UI:

- **Doctor:** Data-heavy, teal-themed editor.
- **Nurse:** Clinical blue, large touch targets (44pt+), high-contrast data display.
- **Admin:** Technical purple, dashboard-focused system metrics.

Accessibility: WCAG AA Compliance, high-contrast ratios, and clinical-standard iconography.



SIMULATION FIDELITY

Flow Logic Engine

The engine simulates volumetric flow using MathML calculations every second:

$$V_{\text{infused}} = V_{\text{prev}} + \frac{R_{\text{infusion}}}{3600}$$

Where R is mL/hr converted to mL/s. The engine automatically transitions to **KVO (Keep Vein Open)** mode when primary volume is depleted.



| CLINICAL INTEGRITY

100%

Traceability of Actions



Immutable Audit Logs

Every clinical action is cryptographically logged with a user ID and timestamp:

- **Protocol Changes:** Old vs. New value capture.
- **Pump Controls:** Start, Pause, Stop, Emergency.
- **Safety Alerts:** Soft limit overrides recorded.
- **Technical:** Auth attempts and biometric success.

PHASE 04

Validation & Assurance

Testing Framework and Requirement Traceability

VALIDATION TRACEABILITY

Traceability Matrix (Extract):

Req ID	Requirement Description	Test Case (Validation)	Status
F-D-02	Mandatory Hard/Soft Limit definition	V-D-02: Input negative/empty values	✓ Pass
F-N-04	Real-time input validation against limits	V-N-02: Rate > Hard Limit (Block Start)	✓ Pass
F-N-06	Emergency Stop functionality	V-N-05: Single tap + 2s hold verification	✓ Pass
SF-N-03	Battery drain simulation logic	V-N-01: Auto-stop at 5% battery level	✓ Pass

Project Conclusion

The Smart Infusion Pump Simulator provides a high-assurance platform for clinical excellence, combining robust architecture with strict safety guardrails.



Technical Inquiry
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Documentation
docs.simpump.io

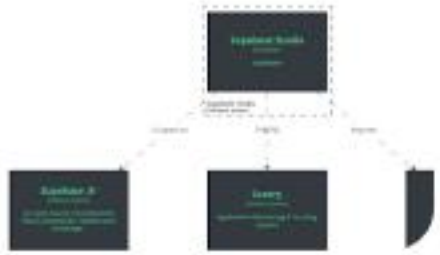
Any Questions?

IMAGE SOURCES



https://img.freepik.com/premium-vector/healthcare-insurance-cyber-security-protection-shield-with-check-mark-icon-inside-sphere-shield_127544-3581.jpg

Source: www.freepik.com



<https://storage.ghost.io/c/03/23/032318a1-e7f3-4abd-83b5-f72b4de9ac77/content/images/2022/07/image-25.png>

Source: www.workingsoftware.dev



<https://www.nursefinders.com/contentassets/44cb9239df59460a8a115dcd829f3059/nurse-checking-for-healthcare-worker-discounts.jpg>

Source: www.nursefinders.com



<https://www.akasinfusions.com/blog/wp-content/uploads/2025/05/infusion-pumps-everything-you-need-to-know.webp>

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